

CUSHING, OK, USA

WMO: 720357

Lat: 35.950N

Lon: 96.773W

Elev: 279

StdP: 98.02

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.4	-15.7	1.0	-8.2	-12.8	1.3	-4.9	11.4	9.2	10.4	9.7	2.4	20	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	38.0	24.5	36.3	24.5	34.9	24.7	26.8	33.9	26.0	33.2	25.5	32.4	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.9	30.4	24.1	19.6	29.5	23.1	18.4	28.3	85.8	33.9	82.2	33.4	79.4	32.4	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.7	-15.1	40.4	4.5	2.8	-18.3	42.4	-21.0	44.0	-23.5	45.6	-26.8	47.6		
(5)			-15.7	27.5	4.2	1.0	-18.7	28.2	-21.2	28.8	-23.6	29.3	-26.6	30.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	3.9	5.5	11.4	15.6	20.4	25.7	27.7	27.4	23.0	16.3	10.2	4.5	(6)	
	DBStd	9.81	5.81	6.42	6.07	5.26	4.34	2.78	2.91	3.24	3.72	5.15	5.25	5.38	(7)	
	HDD10.0	674	202	151	55	13	1	0	0	0	0	9	60	183	(8)	
	HDD18.3	1960	448	359	227	112	30	0	0	0	6	102	248	428	(9)	
	CDD10.0	2873	12	26	98	181	324	471	549	539	389	203	66	13	(10)	
	CDD18.3	1117	0	1	11	29	95	221	292	281	145	38	3	0	(11)	
	CDH23.3	12461	2	14	101	309	886	2417	3556	3284	1464	387	40	1	(12)	
	CDH26.7	5978	0	2	22	77	298	1123	1936	1777	631	109	4	0	(13)	
(14) Wind	WSAvg	3.0	3.2	3.4	3.7	3.9	3.1	3.0	2.4	2.2	2.2	2.8	3.1	3.0	(14)	
(15) Precipitation	PrecAvg	953	32	39	75	96	128	131	82	73	100	84	63	47	(15)	
	PrecMax	1340	100	110	216	207	280	325	190	189	184	253	172	129	(16)	
	PrecMin	727	1	1	13	7	32	41	20	2	22	26	0	2	(17)	
	PrecStd	144	26	30	49	52	68	74	47	45	41	61	54	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	25.3	29.7	32.1	33.0	37.5	41.3	42.5	37.6	32.0	27.3	22.3	(19)	
		MCWB	13.1	15.6	18.0	20.0	22.3	25.2	24.9	24.6	24.1	21.4	17.6	16.2	(20)	
	2%	DB	19.8	21.9	26.2	28.7	31.4	35.2	38.0	38.9	34.0	29.9	24.0	18.5	(21)	
		MCWB	12.6	13.6	16.4	19.2	22.1	24.8	24.6	24.3	23.9	21.2	16.4	14.0	(22)	
	5%	DB	16.3	18.8	23.0	27.0	30.1	33.2	36.3	36.8	32.3	27.5	21.9	16.0	(23)	
		MCWB	10.3	12.7	15.5	18.0	21.8	24.1	24.5	24.6	23.3	19.3	15.1	10.9	(24)	
	10%	DB	13.0	16.2	21.2	25.0	27.9	32.4	34.9	34.8	30.4	25.3	19.0	12.9	(25)	
		MCWB	7.7	11.3	14.7	17.2	21.0	24.0	24.9	24.8	22.5	18.5	14.0	9.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	17.1	19.9	22.1	25.0	27.1	28.1	27.7	26.3	23.9	19.9	18.5	(27)	
		MCDB	20.0	22.0	26.0	27.2	30.4	34.0	34.6	35.1	34.7	29.4	23.8	22.5	(28)	
	2%	WB	13.0	15.4	18.5	20.7	23.7	26.3	26.9	26.5	25.2	22.7	17.9	14.9	(29)	
		MCDB	17.6	20.4	23.9	25.6	28.9	32.8	33.6	34.5	32.2	27.6	21.5	17.1	(30)	
	5%	WB	10.6	13.8	17.2	19.8	22.8	25.6	26.0	25.8	24.3	21.6	16.6	12.1	(31)	
		MCDB	15.2	17.6	22.2	24.5	27.9	31.9	33.1	33.3	30.1	25.1	20.1	14.6	(32)	
	10%	WB	8.4	11.2	15.7	18.7	22.1	24.9	25.4	25.1	23.4	20.1	14.8	9.8	(33)	
		MCDB	13.0	16.3	20.2	23.4	26.8	30.8	32.5	32.3	28.6	23.6	18.5	13.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.1	12.9	12.9	11.4	11.2	12.0	12.0	12.6	13.1	13.1	11.2	(35)	
		MCDBR	17.5	16.4	15.3	15.1	13.2	12.8	14.2	14.9	14.6	14.8	15.5	15.5	(36)	
	5% WB	MCWBR	10.7	9.7	8.3	7.8	6.0	4.9	4.4	4.4	5.9	7.4	9.0	10.3	(37)	
		MCWBR	16.0	14.4	12.7	13.0	11.0	11.2	12.1	12.3	12.2	11.6	13.7	12.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.297	0.303	0.342	0.381	0.415	0.430	0.434	0.433	0.391	0.348	0.316	0.292	(40)	
		taud	2.504	2.502	2.419	2.325	2.298	2.291	2.320	2.264	2.371	2.479	2.508	2.528	(41)	
	Ebn at Noon		908	943	927	902	871	855	848	842	864	878	877	891	(42)	
		Edh at Noon	81	91	109	126	131	132	128	132	113	93	80	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.66	3.27	4.27	5.31	5.81	6.65	6.70	6.06	5.04	3.80	2.81	2.25	(44)	
	RadStd		0.27	0.49	0.27	0.37	0.58	0.63	0.37	0.38	0.44	0.47	0.35	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page